

HealthConnect Clinic Experience Lab

Week 4: Project Summary (Data Analytics Track)

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1. Problem Addressed

As the Data Analytics track, my focus is on understanding the HealthConnect appointment data well enough to explain why patients miss appointments and where the clinic loses the most capacity as a result, laying the groundwork for the deeper analysis and dashboarding planned for later weeks.

2. Resources Used

- HealthConnect_Appointment_Data (5,000 appointment records, 18 variables)
- HealthConnect Data Dictionary (field definitions and expected value ranges)

3. Key Observations from Week 4

- Nearly half of all appointments (48.5%) end in a No-Show, a large enough share to justify focused analysis.
- The dataset is clean and internally consistent: no duplicates, no logical violations between historical fields or dates, and missingness is minimal (under 2%) apart from reminder_channel, which is fully explained by reminder_sent = No.
- waiting_time_minutes is populated even for No-Show and Cancelled appointments with near-identical averages across outcomes, flagged as an anomaly, likely not a genuine attendance-linked measurement.
- An early cross-tab shows previous_no_shows is the most promising signal found so far: patients with a prior no-show have a 55.4% no-show rate on their next appointment vs. 43.5% for those with a clean history.

4. Proposed Approach

Move from this initial quality assessment into Power Query for data cleaning and shaping, then build out the data model and DAX measures in Power BI to calculate the five identified KPIs across relevant segments, shaping the findings toward an interactive dashboard for clinic stakeholders.

5. Key Considerations

- A decision is still needed on how to handle the small number of missing distance_to_clinic_km and waiting_time_minutes values.
- appointment outcome bundles Cancelled and No-Show, which are operationally different behaviors, KPI definitions must treat them separately.
- This is fictional, synthetic data and findings support the learning exercise, not real clinic operations.

6. Proposed Focus for Week 5

Clean and load the data via Power Query, build the Power BI data model, calculate the five candidate KPIs as DAX measures, and produce initial visualizations to confirm which factors most strongly relate to no-shows ahead of full dashboard design.